

Lecture 8: Risk Sensitive Control and Grossman-Zhou Problem

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8.1 The HJB Equation as a Nonlinear Eigenvalue Problem

In traditional discounted portfolio problems (e.g., Merton), investors optimize expected utility over additive consumption streams. However, in long-horizon quantitative asset management, benchmark tracking, and robust portfolio theory (e.g., Bielecki and Pliska [1]), investors often optimize an exponentiated performance criterion over infinite horizons.

This formulation penalizes sample-path variance of cumulative growth, shifting the mathematical structure from linear PDEs to **nonlinear eigenvalue problems**.

8.1.1 Risk-Sensitive Control Formulation

Consider a state process $X_t^u \in \mathbb{R}^n$ driven by controlled drift $\mu(X_t^u, u_t)$, volatility $\sigma(X_t^u, u_t)$, and standard Brownian motion W_t ,

$$dX_t^u = \mu(X_t^u, u_t) dt + \sigma(X_t^u, u_t) dW_t .$$

The running reward rate is $F(X_t^u, u_t)$. Rather than maximizing expected additive reward $\mathbb{E} \left[\int_0^T F dt \right]$, the risk-sensitive investor maximizes the **long-run asymptotic risk-adjusted growth rate**,

$$J(u) = \lim_{T \rightarrow \infty} \frac{1}{\theta T} \ln \mathbb{E} \left[\exp \left(\theta \int_0^T F(X_t^u, u_t) dt \right) \right] , \quad (8.1)$$

where $\theta \neq 0$ is the **risk-sensitivity parameter**:

- **Risk-Averse** ($\theta < 0$): Exponentiating with $\theta < 0$ penalizes negative fluctuations in cumulative rewards, forcing the controller to prefer paths with low variance.
- **Risk-Seeking** ($\theta > 0$): Exponentiating with $\theta > 0$ rewards upside tail volatility.
- **Risk-Neutral** ($\theta \rightarrow 0$): Taking the limit $\theta \rightarrow 0$ recovers standard ergodic expected reward $\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\int_0^T F dt \right]$.

8.1.2 Detailed Derivation of the Stationary PDE

To derive the governing stationary PDE without assuming a prior linear PDE structure, we apply the Multiplicative Dynamic Programming Principle (DPP) over an infinitesimal time step and expand via Itô's Lemma under risk aversion ($\theta < 0$).

Step 1: Dynamic Risk-Sensitive HJB via Multiplicative DPP For $\theta < 0$, define the time-dependent value function

$$V(t, x) := \sup_{u \in \mathcal{A}} \frac{1}{\theta} \ln \mathbb{E}_{t,x} \left[\exp \left(\theta \int_t^T F(X_s^u, u_s) ds \right) \right]. \quad (8.2)$$

Multiplying $V(t, x)$ by $\theta < 0$ reverses the inequality, turning the supremum into an infimum. Exponentiating both sides yields the multiplicative expectation form

$$e^{\theta V(t,x)} = \inf_{u \in \mathcal{A}} \mathbb{E}_{t,x} \left[\exp \left(\theta \int_t^T F(X_s^u, u_s) ds \right) \right]. \quad (8.3)$$

By the Dynamic Programming Principle over $[t, t+h]$ for a small time step $h > 0$,

$$e^{\theta V(t,x)} = \inf_{u \in \mathcal{A}} \mathbb{E}_{t,x} \left[\exp \left(\theta \int_t^{t+h} F(X_s^u, u_s) ds \right) e^{\theta V(t+h, X_{t+h}^u)} \right]. \quad (8.4)$$

For small h , expanding the running integral exponential gives

$$\exp \left(\theta \int_t^{t+h} F(X_s^u, u_s) ds \right) = 1 + \theta F(x, u)h + o(h). \quad (8.5)$$

Next, apply Itô's Lemma directly to $g(t, x) = e^{\theta V(t,x)}$,

$$\partial_t (e^{\theta V}) = \theta e^{\theta V} \partial_t V, \quad (8.6)$$

$$\frac{\partial}{\partial x} (e^{\theta V}) = \theta e^{\theta V} \frac{\partial V}{\partial x}, \quad (8.7)$$

$$\frac{\partial^2}{\partial x^2} (e^{\theta V}) = \theta e^{\theta V} \frac{\partial^2 V}{\partial x^2} + \theta^2 e^{\theta V} \left(\frac{\partial V}{\partial x} \right)^2. \quad (8.8)$$

The infinitesimal generator $\mathcal{A}^u$ of $g(t, x) = e^{\theta V(t,x)}$ is

$$\begin{aligned} \mathcal{A}^u (e^{\theta V}) &:= \partial_t (e^{\theta V}) + \mu(x, u) \frac{\partial}{\partial x} (e^{\theta V}) + \frac{1}{2} \sigma^2(x, u) \frac{\partial^2}{\partial x^2} (e^{\theta V}) \\ &= \theta e^{\theta V(t,x)} \left[\partial_t V + \mathcal{L}^u V(t, x) + \frac{\theta}{2} \sigma^2(x, u) \left(\frac{\partial V}{\partial x} \right)^2 \right], \end{aligned} \quad (8.9)$$

where $\mathcal{L}^u V := \mu(x, u) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2(x, u) \frac{\partial^2 V}{\partial x^2}$ is the standard generator.

Substituting these expansions into the dynamic programming equation,

$$e^{\theta V(t,x)} = \inf_{u \in \mathcal{A}_x} \mathbb{E}_{t,x} \left[\left(1 + \theta F(x, u)h + o(h) \right) \left(e^{\theta V(t,x)} + \mathcal{A}^u (e^{\theta V}) h + o(h) \right) \right]. \quad (8.10)$$

Subtracting $e^{\theta V(t,x)}$ from both sides yields

$$0 = \inf_{u \in \mathcal{A}_x} \left\{ \theta e^{\theta V(t,x)} \left[\partial_t V + \mathcal{L}^u V(t, x) + \frac{\theta}{2} \sigma^2(x, u) \left(\frac{\partial V}{\partial x} \right)^2 + F(x, u) \right] h \right\} + o(h). \quad (8.11)$$

Dividing through by $\theta e^{\theta V(t,x)}h$ and taking $h \rightarrow 0$: because $\theta < 0$, dividing by a negative quantity reverses the inf back into a sup, yielding the dynamic risk-sensitive HJB equation

$$\partial_t V(t, x) + \sup_{u \in \mathcal{A}_x} \left\{ \mathcal{L}^u V(t, x) + \frac{\theta}{2} \sigma^2(x, u) \left(\frac{\partial}{\partial x} V(t, x) \right)^2 + F(x, u) \right\} = 0 . \quad (8.12)$$

Step 2: Separation of Variables (Asymptotic Growth Ansatz) As $T \rightarrow \infty$, explicit dependence on the remaining horizon $T - t$ stabilizes into a constant long-run growth rate $\lambda \in \mathbb{R}$. We propose the additive time-separable ansatz

$$V(t, x) = \lambda(T - t) + w(x) . \quad (8.13)$$

Step 3: Substituting Derivatives into the HJB Calculating partial derivatives ($\partial_t V = -\lambda$, $\partial_x V = w'(x)$, and $\partial_{xx} V = w''(x)$) and substituting into (8.12) yields

$$-\lambda + \sup_{u \in \mathcal{A}_x} \left\{ \mathcal{L}^u w(x) + \frac{\theta}{2} \sigma^2(x, u) (w'(x))^2 + F(x, u) \right\} = 0 . \quad (8.14)$$

Rearranging gives the stationary **Nonlinear Eigenvalue PDE**

$$\sup_{u \in \mathcal{A}_x} \left\{ \mathcal{L}^u w(x) + \frac{\theta}{2} \sigma^2(x, u) (w'(x))^2 + F(x, u) \right\} = \lambda . \quad (8.15)$$

8.1.3 Structural and Financial Interpretation

1. **Nonlinear Eigenvalue** ($\lambda \in \mathbb{R}$): Acts as the scalar eigenvalue of the non-linear differential operator. Economically, λ represents the annualized **certainty-equivalent growth rate** of the portfolio.
2. **Eigenfunction** ($w(x)$): Acts as the spatial eigenfunction, measuring the transient, relative valuation of being in state x compared to steady-state growth.
3. **Quadratic Gradient Penalty Term** ($\frac{\theta}{2} \sigma^2(w')^2$): When $\theta < 0$, the quadratic term $\frac{\theta}{2} \sigma^2(w')^2 \leq 0$ acts as a penalty on state variance. In financial engineering, this connects risk-sensitive control directly to **Robust / Worst-Case Control**,

$$\frac{\theta}{2} \sigma^2(x, u) (w'(x))^2 = \min_v \left\{ v \cdot \sigma(x, u) w'(x) - \frac{1}{2\theta} v^2 \right\} . \quad (8.16)$$

The investor acts as if playing a zero-sum differential game against a hostile "Nature" that actively distorts market drift by v .

8.2 Grossman-Zhou (1993): Portfolio Optimization under Draw-down Constraints

In practical asset management, fund managers face strict capital preservation mandates: wealth must never fall below a fixed percentage of its historical peak (high-water mark). Grossman and Zhou [3] formulate this drawdown-constrained problem as an infinite-horizon ergodic growth problem.

8.2.1 Mathematical Objective and Pathwise Constraint Set

Let X_t^π denote portfolio wealth under a dollar allocation strategy π_t in a market with interest rate r , drift μ , and volatility σ . Let M_t be the running historical maximum (high-water mark) of wealth up to time t ,

$$M_t := \max_{0 \leq s \leq t} X_s^\pi . \quad (8.17)$$

For power utility exponent $\delta \in (0, 1)$ (relative risk aversion $1 - \delta$), the investor maximizes the asymptotic growth eigenvalue λ^* ,

$$\lambda^* = \sup_{\pi \in \mathcal{A}} \lim_{T \rightarrow \infty} \frac{1}{T} \ln \mathbb{E} \left[\frac{(X_T^\pi)^\delta}{\delta} \right] , \quad (8.18)$$

subject to the continuous, pathwise drawdown constraint parameterized by floor fraction $\alpha \in (0, 1)$,

$$\mathbb{P} \left(X_t^\pi \geq \alpha M_t \quad \forall t \geq 0 \right) = 1 \quad \implies \quad \text{Maximum Drawdown} \leq 1 - \alpha . \quad (8.19)$$

8.2.2 Detailed Derivation of Linear Scale Invariance and Dimensional Reduction

The state space initially appears two-dimensional: $(X_t, M_t) \in \mathcal{D} = \left\{ (x, m) \in \mathbb{R}_+^2 \mid \alpha m \leq x \leq m \right\}$. We now prove that scale invariance collapses $\mathcal{D}$ to a compact 1D interval.

Step 1: Scale Invariance of Market Dynamics Consider scaling initial wealth x and initial high-water mark m by an arbitrary constant scaling factor $k > 0$,

$$(x, m) \longrightarrow (kx, km) . \quad (8.20)$$

Under linear wealth dynamics $dX_t^\pi = (rX_t^\pi + \pi_t(\mu - r))dt + \sigma\pi_t dW_t$, scaling the dollar strategy $\pi_t \rightarrow k\pi_t$ yields

$$dX_t^{k\pi} = k dX_t^\pi \implies X_t^{k\pi} = kX_t^\pi . \quad (8.21)$$

Step 2: Scale Invariance of Peak Wealth and Constraints The running high-water mark scales identically,

$$M_t^{k\pi} = \max_{0 \leq s \leq t} (kX_s^\pi) = k \max_{0 \leq s \leq t} X_s^\pi = kM_t^\pi . \quad (8.22)$$

Scaling both sides of the drawdown inequality by $k > 0$ preserves admissibility,

$$X_t^{k\pi} \geq \alpha M_t^{k\pi} \iff kX_t^\pi \geq k\alpha M_t^\pi \iff X_t^\pi \geq \alpha M_t^\pi . \quad (8.23)$$

Step 3: Degree- δ Homogeneity of the Value Function Because power utility satisfies $U(kX_T) = \frac{(kX_T)^\delta}{\delta} = k^\delta U(X_T)$, the time-dependent value function $V(t, x, m)$ satisfies degree- δ spatial homogeneity,

$$V(t, kx, km) = k^\delta V(t, x, m) \quad \forall k > 0 . \quad (8.24)$$

Step 4: Dimensional Reduction and Definition of $\phi(z)$ Define the normalized state ratio variable $Z_t := \frac{X_t}{M_t} \in [\alpha, 1]$. Assuming a time-separable growth ansatz $V(t, x, m) = e^{\lambda t} v(x, m)$ and choosing $k = \frac{1}{m} > 0$ to normalize peak wealth to 1,

$$V(t, x, m) = e^{\lambda t} v \left(m \cdot \frac{x}{m}, m \cdot 1 \right) = e^{\lambda t} m^\delta v \left(\frac{x}{m}, 1 \right) . \quad (8.25)$$

We formally define the reduced 1D valuation function (eigenfunction) as $\phi(z) := v(z, 1)$ where $z = x/m$. The value function factors into

$$V(t, x, m) = e^{\lambda t} m^\delta \phi(z) , \quad z \in [\alpha, 1] . \quad (8.26)$$

Step 5: Compact Domain Bounds The physical state bounds dictate the domain of Z_t :

1. **Upper Bound** ($z \leq 1$): Wealth X_t cannot exceed its running historical peak M_t , forcing $Z_t = \frac{X_t}{M_t} \leq 1$.
2. **Lower Bound** ($z \geq \alpha$): The drawdown floor enforces $X_t \geq \alpha M_t$, forcing $Z_t = \frac{X_t}{M_t} \geq \alpha$.

Thus, the unbounded 2D wedge domain collapses to the compact 1D interval $Z_t \in [\alpha, 1]$.

8.2.3 Boundary Value Problem on the Reduced State Space

The compact domain $Z_t \in [\alpha, 1]$ dictates two physical boundary conditions at the domain extremes, which together with the interior HJB equation form a complete boundary value problem (BVP).

Boundary	Mathematical Formulation	Economic / Mathematical Reasoning
Drawdown Floor ($z = \alpha$)	$\phi(\alpha) = 0 \implies \pi^*(\alpha) = 0$	Volatility must vanish to guarantee $X_t \geq \alpha M_t$ almost surely.
High-Water Mark ($z = 1$)	$\phi'(1) = \delta\phi(1)$	Neumann reflection condition for singular control on M_t .

1. Physical Boundary Conditions on $[\alpha, 1]$:

- **At the Drawdown Floor** ($Z_t = \alpha$): To guarantee $X_t \geq \alpha M_t$ pathwise, risky asset allocation must vanish as wealth approaches the floor, forcing the value function to zero at the boundary,

$$\phi(\alpha) = 0 \implies \lim_{Z_t \rightarrow \alpha} \pi_t^* = 0 . \quad (8.27)$$

At $z = \alpha$, portfolio wealth earns purely the risk-free rate r , preventing any possibility of breaching the drawdown limit.

- **At the High-Water Mark** ($Z_t = 1$): Peak wealth M_t increases only when $X_t = M_t$ ($Z_t = 1$). Because M_t is a non-decreasing process acting only at $x = m$, the marginal value of pushing up the peak m must be zero at reflection,

$$\left. \frac{\partial V}{\partial m}(t, x, m) \right|_{x=m} = 0 . \quad (8.28)$$

Differentiating $V(t, x, m) = e^{\lambda t} m^\delta \phi(x/m)$ with respect to m via the product rule,

$$\frac{\partial V}{\partial m} = e^{\lambda t} \delta m^{\delta-1} \phi\left(\frac{x}{m}\right) + e^{\lambda t} m^\delta \phi'\left(\frac{x}{m}\right) \left(-\frac{x}{m^2}\right) = e^{\lambda t} m^{\delta-1} \left(\delta\phi(z) - z\phi'(z)\right) . \quad (8.29)$$

Setting $z = x/m = 1$ yields the **Neumann Boundary Condition**

$$\phi'(1) = \delta\phi(1) . \quad (8.30)$$

2. Reduced HJB ODE in the Interior ($z \in (\alpha, 1)$): In the interior, wealth is strictly between the floor and the peak ($X_t \in (\alpha M_t, M_t)$), so $dM_t = 0$. Defining the peak-scaled dollar strategy $y_t := \frac{\pi_t}{M_t}$, the reduced function $\phi(z)$ satisfies the stationary HJB equation

$$\sup_y \left\{ \frac{1}{2} \sigma^2 y^2 \phi''(z) + (rz + y(\mu - r)) \phi'(z) \right\} = \lambda \phi(z) . \quad (8.31)$$

Taking the first-order condition (FOC) with respect to y gives the optimal allocation policy $y^*(z)$,

$$y^*(z) = -\frac{\mu - r}{\sigma^2} \frac{\phi'(z)}{\phi''(z)} . \quad (8.32)$$

Substituting $y^*(z)$ back into the HJB operator completes the non-linear second-order ordinary differential equation

$$\lambda\phi(z) - rz\phi'(z) + \frac{(\mu - r)^2}{2\sigma^2} \frac{(\phi'(z))^2}{\phi''(z)} = 0, \quad z \in (\alpha, 1) , \quad (8.33)$$

subject to the boundary conditions $\phi(\alpha) = 0$ and $\phi'(1) = \delta\phi(1)$.

3. Analytical Solution via Wealth Cushion Ansatz: To satisfy $\phi(\alpha) = 0$ while maintaining scale-invariance, we propose the wealth cushion power ansatz

$$\phi(z) = C(z - \alpha)^p, \quad C > 0, \quad p > 0 . \quad (8.34)$$

Differentiating $\phi(z)$ with respect to z ,

$$\phi'(z) = Cp(z - \alpha)^{p-1} , \quad (8.35)$$

$$\phi''(z) = Cp(p-1)(z - \alpha)^{p-2} . \quad (8.36)$$

The risk-aversion ratio $\frac{\phi'(z)}{\phi''(z)}$ simplifies to

$$\frac{\phi'(z)}{\phi''(z)} = \frac{z - \alpha}{p - 1} = -\frac{z - \alpha}{1 - p} . \quad (8.37)$$

Substituting this ratio into the high-water mark boundary condition $\phi'(1) = \delta\phi(1)$ uniquely fixes the power exponent p ,

$$Cp(1 - \alpha)^{p-1} = \delta C(1 - \alpha)^p \implies p = \delta(1 - \alpha) . \quad (8.38)$$

Working in discounted wealth coordinates (absorbing the risk-free drift r), substituting $\frac{(\phi'(z))^2}{\phi''(z)} = -\frac{p}{1-p}(z - \alpha)^p$ into equation (8.33) and dividing through by $(z - \alpha)^p$ yields the long-run growth eigenvalue λ ,

$$\lambda = \frac{(\mu - r)^2}{2\sigma^2} \left(\frac{p}{1 - p} \right) = \frac{(\mu - r)^2}{2\sigma^2} \left(\frac{\delta(1 - \alpha)}{1 - \delta(1 - \alpha)} \right) . \quad (8.39)$$

8.2.4 Core Structural Result: Constant Cushion Allocation

Solving the stationary HJB ODE for $\phi(z)$ on $z \in [\alpha, 1]$ subject to these boundary conditions yields the central economic result of Grossman and Zhou (1993),

$$\pi_t^* = \kappa \cdot \underbrace{(X_t - \alpha M_t)}_{\text{Cushion } C_t} , \quad (8.40)$$

where the constant multiplier $\kappa > 0$ is given by

$$\kappa = \frac{\mu - r}{\sigma^2(1 - p)} = \frac{\mu - r}{\sigma^2[1 - \delta(1 - \alpha)]} . \quad (8.41)$$

Economic Intuition of Cushion Strategy:

- **The Wealth Cushion** ($C_t := X_t - \alpha M_t$): The investor allocates risk strictly on surplus wealth above the floor C_t , rather than total nominal wealth X_t .
- **Endogenous Risk Aversion Boost:** The drawdown constraint induces an *effective relative risk aversion* $\gamma_{\text{eff}} := 1 - p = 1 - \delta(1 - \alpha)$. Because $p = \delta(1 - \alpha) < \delta$, the constraint strictly increases effective risk aversion, forcing a more conservative portfolio multiplier ($\kappa < \kappa_{\text{Merton}}$).
- **Automatic Capital Preservation:** As wealth declines toward the floor ($X_t \rightarrow \alpha M_t$), the cushion vanishes ($C_t \rightarrow 0$). This automatically scales risky exposure down to zero ($\pi_t^* \rightarrow 0$), guaranteeing capital preservation without triggering catastrophic liquidation.
- **Unconstrained Limit** ($\alpha \rightarrow 0$): As the floor fraction vanishes ($\alpha \rightarrow 0$), $p \rightarrow \delta$, $C_t \rightarrow X_t$, and $\gamma_{\text{eff}} \rightarrow 1 - \delta$, recovering Merton's classical power-utility policy $\pi_t^* = \left(\frac{\mu - r}{(1 - \delta)\sigma^2} \right) X_t$.

8.3 Further Reading

Advanced Extensions: Students interested in generalizations of this benchmark result should consult Cvitanić and Karatzas [2], who extended the Grossman-Zhou framework to multi-asset markets with time-varying parameters using martingale and convex duality methods.

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References

- [1] T. R. Bielecki and S. R. Pliska. Risk-sensitive dynamic asset management. *Applied Mathematics and Optimization*, 39(3):337–360, 1999.
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- [3] S. J. Grossman and Z. Zhou. Optimal investment strategies for controlling drawdowns. *Mathematical finance*, 3(3):241–276, 1993.